

EXPERIMENT - 5

5. Write a Python function to analyze the histogram of the given input image based on color levels using Open CV.

Aim:

To write a Python program using OpenCV to analyze the histogram of a given input image based on its color levels.

Procedure:

1. Import the required libraries cv2 and matplotlib.
2. Read the input image using cv2.imread().
3. Convert the image from BGR to RGB format for displaying.
4. Calculate the histogram for the Red, Green, and Blue color channels using cv2.calcHist().
5. Display the input image.
6. Plot the histograms of the three color channels.
7. Observe the distribution of color intensity levels from 0 to 255.

Program:

```
import cv2
import matplotlib.pyplot as plt

img = cv2.imread("input.jpg")

img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

colors = ('b', 'g', 'r')

plt.figure(figsize=(10, 5))

for i, color in enumerate(colors):
    hist = cv2.calcHist([img], [i], None, [256], [0, 256])
    plt.plot(hist, color=color)
```

```
plt.title("Color Histogram")  
plt.xlabel("Color Intensity Level")  
plt.ylabel("Number of Pixels")  
plt.xlim([0, 256])  
plt.grid()
```

```
plt.show()
```

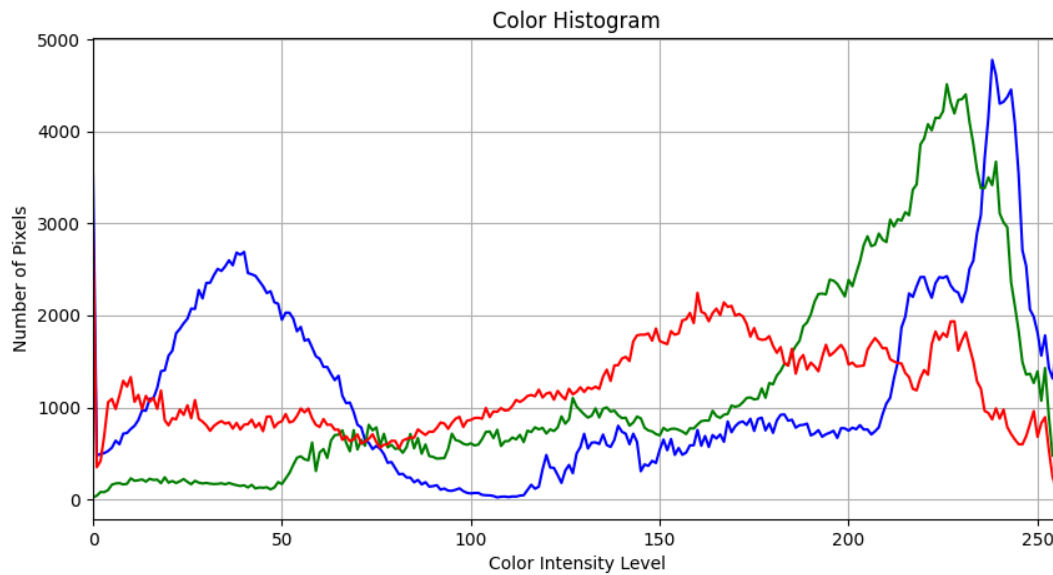
```
plt.imshow(img_rgb)  
plt.title("Input Image")  
plt.axis("off")  
plt.show()
```

Sample Input:

Input.jpg



Sample Output:



Result

Thus, the Python program successfully analyzed the histogram of the given input image based on Red, Green, and Blue color levels using OpenCV.